Perfect Completeness for 2-to-1 Games

An expository account of the proof

A rewrite, for human readers, of the manuscript *Perfect completeness for 2-to-1 games* (OpenAI, September 23, 2026)

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Abstract

In September 2026 OpenAI posted a 53-page manuscript, written by an unreleased model, proving Khot's 2-to-1 Games Conjecture with perfect completeness. For every $\delta > 0$ it is NP-hard to tell a satisfiable 2-to-1 game from one in which no labeling satisfies more than a δ fraction of the constraints. The near-perfect version, with completeness $1 - \varepsilon$, was the 2-to-2 Games Theorem of Khot, Minzer and Safra. Perfect completeness needs a new construction, because the standard tool for driving soundness down, parallel repetition, destroys the 2-to-1 structure. The manuscript is complete but compressed. This note explains the proof for a complexity theorist who knows the PCP theorem and the outline of the 2-to-2 theorem. It covers the construction (a tree of function spaces, with questions given by unlabeled partitions) and the three soundness steps: an upper decoder built on the Grassmann inverse theorem, a passage from "rank one on the tested rows" to bounded full rank, and a final appeal to parallel repetition on "clean" subtrees. Elementary steps are proved in full. For the rest, the note explains what is proved and points to the manuscript. The result has a Lean formalization in the same release.

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1 Introduction

1.1 The theorem

A *projection game* has a bipartite graph between question sets U and V, an answer set at each question, and on each edge $e = (u, v)$ a map π_e from answers at u to answers at v. A labeling satisfies e when the right answer is the image of the left one, and the *value* is the largest satisfiable fraction of edges. In a *2-to-1 game* the left alphabet is $[2q]$, the right alphabet is $[q]$, and every map is exactly two-to-one: $|\pi_e^{-1}(b)| = 2$ for every right answer b.

Theorem 1.1 (Theorem 1.1 of [1])**.** *For every fixed rational $\delta \in (0, 1)$ there are $q = q(\delta)$ and a deterministic polynomial-time reduction from 3SAT to 2-to-1 games on alphabets $[2q]$ and $[q]$ such that satisfiable formulas give games of value 1 and unsatisfiable formulas give games of value at most δ. The output is an explicit list of unweighted edges with explicit exact-two projection tables.*

This is Khot's 2-to-1 Games Conjecture [14] in its strongest form. The phrase that matters is *value* 1: the YES case is *perfectly* satisfiable. That is what separates it from the Unique Games Conjecture, which is false with perfect completeness, and from the 2-to-2 Games Theorem of Khot, Minzer and Safra, which has completeness $1 - \varepsilon$ [15, 7, 3, 16].

1.2 Why perfect completeness matters, and why it is hard

Perfect completeness is the hypothesis behind hardness results for *satisfiable* problems: coloring graphs that really are 3- or 4-colorable, satisfiable CSPs, and similar problems [5, 11, 12, 18]. A reduction that starts from a game with completeness $1 - \varepsilon$ produces YES instances that are only almost colorable. The manuscript derives three consequences:

- **Maximum k-colorable subgraph** (Corollary 1.2). For large fixed k it is NP-hard to tell a k-colorable graph from one in which every k-coloring properly colors at most a $1 - 1/k + O(\log k/k^2)$ fraction of the edges. This had previously been known only conditionally, via Guruswami and Sinop [11].
- **Satisfiable Not-Two** (Corollary 1.3), through O'Donnell and Wu [18]. This recovers Håstad's unconditional theorem [13]: a three-query PCP with perfect completeness and soundness $5/8 + \varepsilon$. The manuscript notes that the same corollary also follows from the Fei–Minzer–Wang 4-to-1 theorem.
- **Coloring 3-colorable graphs with any constant number of colors**, through Guruswami and Sandeep [12]. Fei, Minzer and Wang had proved this nine days earlier from their perfect-completeness 4-to-1 theorem [9].

Two obstacles explain why this is harder than the 2-to-2 theorem.

> **Obstacle 1: no repetition.** Normally one proves a constant gap and then drives soundness to δ by parallel repetition. But the k-fold repetition of an exactly 2-to-1 projection is 2^k-to-1. Whatever amplification the reduction needs must be built into the 2-to-1 game itself. Repetition is still used, but only on an inner *source* game whose answers are hidden inside a larger structure.

> **Obstacle 2: no slack in the YES case.** Completeness $1 - \varepsilon$ does not imply completeness 1 for any $\varepsilon > 0$, because a family of reductions with completeness approaching 1 need not produce a single satisfiable game. Every test therefore has to be passed *exactly* by an honest assignment. That pushes the construction toward tests defined by algebraic identities rather than by approximate ones.

Prior work. Austrin, O'Donnell, Tan and Wright proved perfect-completeness 2-to-1 hardness with soundness $23/24 + \varepsilon$ [2]. The problem was to get soundness arbitrarily small while keeping both perfect completeness and fibers of size two. In September 2026 Fei, Minzer and Wang posted the corresponding theorem for 4-to-1 games [9], and the manuscript connects it to two ingredients. Its low-rank "Hadamard bucket" viewpoint is inspired by Golowich's construction [10] as mediated by that paper. And its passage from agreement of restricted bilinear forms to a single form of bounded rank has, in the manuscript's words, a "close counterpart" there.

1.3 The idea in one paragraph

An honest prover holds a satisfying assignment x to a small source game, placed at the leaves of a tree. The verifier shows the prover *functions* of x, namely lists of polynomials of growing degree built up the tree, and asks for their values at x. To make the question independent of how these functions are named, the left prover receives only the partition of the answer space according to the joint values, and answers with a part. The right prover receives the same partition with one output direction at one tree node erased, which merges at most two parts. That is the source of the 2-to-1 constraint, and an honest x satisfies it exactly. Joint answers obey exact identities: the answer to gh must be the product of the answers to g and h. A labeling that wins often must therefore look, at some node, like evaluation by a functional z satisfying $z(gh) = z(g)z(h)$ on many tested pairs. Most of the soundness proof shows that, with positive probability, the product form $(g, h) \mapsto z(gh)$ then has bounded rank. A bounded-rank form decodes to a short list of source answers. A final independence argument turns any success into a strategy for many independent copies of the source game, which parallel repetition forbids.

1.4 The proof on one page

1. **Source game** (Section 2.1). Dinur's PCP and Dinur–Steurer parallel repetition give a clause-versus-variable game G_{src} with perfect completeness and value at most σ on NO instances, where σ is a small constant chosen last.
2. **Tree of function spaces** (Section 2.2). A tree of depth d has a copy of a left source question at each leaf. Each leaf carries all Boolean functions of its answer. At an internal node D the space H_D is the sum over children C of the spans of products gh with $g, h \in H_C$. A product of two functions from a descendant lies in every space above it.
3. **Questions are partitions** (Section 2.3). At each internal node the verifier displays an *array* of ℓ_p functions. The question is the unordered partition of the answer space by the joint values

of all the arrays, after discarding unused coordinates. A legal answer is a part. Erasing one output direction at one node merges at most two parts. That gives the 2-to-1 constraint, and a satisfying source assignment passes every test (Proposition 2.5).

4. **Exact identities** (Section 2.5). Shifting a row by products of lower rows does not change the partition. Every labeling therefore has its answers transform by the exact product rule. This is "folding for products", and it shows that a joint answer induces a normalized product form whose tested Gram matrix is yy^{t}, of rank one.
5. **Sampling law** (Sections 2.6 and 3.2). Arrays are sampled along a hidden random root-to-leaf path by a recursive product sampler. Large branching hides the path child, so the cut looks uniform. A sparse random set of children is replaced by projected (variable) questions without changing the law much. Products give small bias independent of dimension.
6. **Upper decoder** (Section 4). Success forces two levels $i < j$ to succeed simultaneously. The Grassmann/shortcode inverse theorem at level j finds a response that is constant on an affine slice. Joint legality turns it into a coset of normalized functionals z whose product forms have tested rank ≤ 1 on the level-i rows. A Fourier heavy-list argument gives the right player a matching guess with probability $\gamma_j > 0$, independent of the lower row count.
7. **Tested rank to full rank** (Section 5). A fixed form of high rank almost never has a rank-≤ 1 Gram matrix on random rows, even on an affine slice. A second application of the inverse theorem therefore shows that a positive part of the agreement comes from forms of bounded full rank s.
8. **Clean children and repetition** (Section 6). A bounded-rank normalized product form gives an odd list of at most s answers at each leaf, and agreement gives the parity pushforward on projected leaves. On "clean" projected children (all contributions zero, with probability exactly independent of the questions there), both players can rebuild their inputs from their own questions. Many clean children behave like independent copies of the source game, and parallel repetition gives a contradiction.
9. **Assembly** (Section 7). The parameters are chosen in a careful order: row counts top down, then the source power, then branching bottom up. Each fiber is then padded to exactly two, and the weights are rounded to an unweighted multiset.

1.5 Status

The manuscript is 53 pages. The repository lists a Lean formalization of the main theorem: a Comparator statement `PerfectCompleteness.lean` formalizing the reduction from binary 3SAT, with value exactly 1 on satisfiable inputs and at most δ otherwise. The check permits only Lean's standard axioms, so no theorem enters the formal proof as an assumption. The Lean development (about 580 files) proves its own versions of the ingredients the paper cites, and its inverse theorem comes from the release's Unique Games library rather than from the Khot–Minzer–Safra paper. I have not run the check. On paper, the proof uses three external theorems as black boxes: Dinur's PCP with perfect completeness [6], the Dinur–Steurer parallel repetition bound [8], and the Khot–Minzer–Safra Grassmann expansion theorem [16]. Section 8 lists what to look at first. Statement numbers such as "Lemma 5.3" refer to the compiled manuscript.

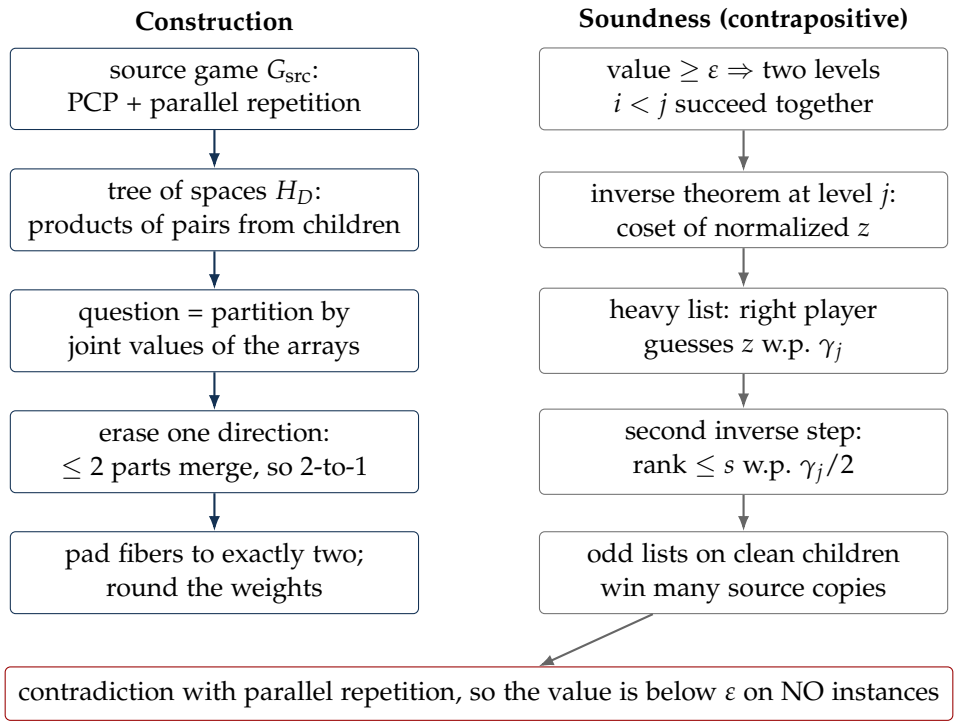


Figure 1: Architecture of the proof.

2 The construction

Throughout, $\mathbb{F}_2$ is the two-element field and all vector spaces are over $\mathbb{F}_2$.

2.1 The source game

Dinur's PCP with perfect completeness [6], followed by routine local encodings, maps a 3CNF formula to a 3CNF formula Ψ with clauses on three distinct variables. If the input is satisfiable, so is Ψ; if not, every assignment violates a γ_0 fraction of clauses (Theorem 2.1). The *clause-versus-variable game* G_Ψ asks the left player a clause c and the right player one of its variables, chosen at random. The left answer lies in Ω_c, the seven satisfying assignments of c; the right answer is a bit; and the constraint checks agreement. Its value is at most $1 - \gamma_0/3$ on NO instances. The *source game* is the t-fold power

$$G_{\text{src}} = G_\Psi^{\otimes t},$$

whose value is at most $(1 - \gamma_0^2/144)^t$ by Dinur–Steurer [8] (Theorem 2.2). The power t is fixed late in the parameter order. Answer sets have sizes 7^t and 2^t.

One small fact is used repeatedly. Each Ω_c projects onto all of $\mathbb{F}_2$ in any single coordinate and onto all of $\mathbb{F}_2^2$ in any two coordinates, since a clause excludes only one of eight triples. *Pair-surjectivity* makes the canonical form of a question below well-defined.

2.2 A tree of function spaces

Fix a depth d, branching numbers $n_1, \ldots, n_d$, row counts $\ell_1, \ldots, \ell_d$ and product counts $R_1, \ldots, R_d$. Let $\mathcal{T}$ be the rooted tree with root at level d, leaves at level 0, and n_p children at every level-p node. A tuple E of left source questions, one per leaf, gives each leaf C an answer domain $\Omega_C(E)$, a product of t clause domains. For a node D, $\Omega_D(E)$ is the product of the leaf domains below it. Different leaves are *always* separate coordinates, even when they carry the same clause.

Definition 2.1 (Definition 3.1)**.** At a leaf C, H_C is the space of *all* functions $\Omega_C \to \mathbb{F}_2$. For any node C let $H_C^2 = \mathrm{span}\{gh : g, h \in H_C\}$ (pointwise products). At an internal node,

$$H_D = \sum_{C \text{ child of } D} H_C^2 \;\subseteq\; \mathbb{F}_2^{\Omega_D},$$

where a function on a subtree is extended by ignoring the other coordinates.

Example 2.2 (What the spaces look like)**.** A leaf space is closed under products, so at level 1 the space H_D consists of sums of functions each depending on a single leaf. At level 2, products of two such functions from one level-1 child depend on at most two leaves under that child. In general every element of H_D at level p is a sum of products of at most 2^{p-1} single-leaf functions. Each product follows a nested binary pattern: at every level only two pieces are multiplied, each coming from a single child. So H_D is *contained in*, but generally smaller than, the span of all such products. Degree doubles at each level, but every term stays inside a single child of D. That locality is what the sampler and the clean-children argument exploit.

The basic properties (Lemma 3.2) are immediate: every space contains 1, $H_C \subseteq H_C^2$, and

$$H_C^2 \subseteq H_D \qquad \text{for every proper descendant } C \text{ of } D. \tag{1}$$

Equation (1) is the structural heart of the construction. *A product of two functions displayed at a descendant is available at every ancestor.*

Projected questions. The analysis also uses *mixed* tuples O in which some leaf clause questions are replaced by their variable questions along source edges. The coordinate projection $\pi : \Omega_D(E) \to \Omega_D(O)$ is surjective. Pullback along it therefore embeds $H_D(O)$ into $H_D(E)$, preserving pointwise products. We identify $H_D(O)$ with its image.

2.3 Arrays, joint values, and questions as partitions

At each internal node D at level p the verifier chooses an *array* $M_D \in K_p \otimes H_D$, with $K_p = \mathbb{F}_2^{\ell_p}$. An array is a list of ℓ_p functions in H_D, called its *rows*. On an assignment $x \in \Omega(E)$ to all leaves, the arrays give the joint value

$$\Psi_{E,\mathbf{M}}(x) = \big(M_D(x)\big)_{D \text{ internal}} \in \prod_D K_{\mathrm{level}(D)}.$$

The honest prover wants to answer with $\Psi(x)$ for a satisfying assignment x. The question is how to pose this so that the prover cannot exploit the names of the functions, and so that left and right questions can be shared across different representations. The manuscript's answer is to pose the question as an *unlabeled partition*.

Definition 2.3 (Canonical key; Lemma 3.3)**.** For a function f on a product domain, discard every coordinate (leaf slot) that f ignores. Replace a clause slot by a single bit u_j whenever f depends on that slot only through one fixed position j, uniformly in all other arguments. The resulting reduced domain carries typed coordinates: full clause slots with their clause ID, and bit slots with their variable ID. The *key* of f consists of a side tag (left or right), these typed coordinates, and the *unordered* partition $\mathcal{P}_f$ of the reduced domain into the nonempty fibers of f. A legal answer is a part $P \in \mathcal{P}_f$. Its *restored output* $\mathrm{out}_f(P)$ is the common value of f on P.

Pair-surjectivity makes the reduced position unique when it exists: a slot that factors through two positions is constant, hence unused. The key depends only on the equivalence relation $f(x) = f(x')$, not on names of output values. Two consequences drive everything.

- **Sharing** (Lemma 3.4). If f is defined on a projected domain $\Omega(O)$, then f and its pullback $f \circ \pi$ have *literally the same key*. A projected slot that f uses becomes, for $f \circ \pi$, a clause slot reducible to the projected position. So a question built on variable questions is the same vertex as the corresponding question on clause questions. This identification lets the analysis switch between E and O without the prover noticing.
- **Joint legality.** Every legal answer is a part, so a single assignment x realizes *all* of the restored block outputs at once. A prover cannot give mutually inconsistent answers about different arrays.

The global quantifier in the reduction rule matters. The function $f(x, b) = (1 + b)x_1 + bx_2$ on $\Omega_c \times \mathbb{F}_2$ reads one clause position for each fixed b, but no single position for all b, so its clause slot is kept in full.

2.4 The one-bit test and perfect completeness

For an internal node D at level p and a nonzero $a \in K_p$, let $Q_{D,a}$ be the quotient $K_p \to K_p/\langle a \rangle$ at block D, and the identity elsewhere. The *edge* joins the left key of $f = \Psi_{E,\mathbf{M}}$ to the right key of $g = Q_{D,a} \circ f$, with map

$$\pi_{f,D,a}(P) = \mathrm{out}_g^{-1}\big(Q_{D,a}\ \mathrm{out}_f(P)\big).$$

Lemma 2.4 (Lemma 3.5)**.** *Each such map is a surjection with at most two preimages per right label. For a fixed labeling, the edge accepts exactly when the restored outputs agree through $Q_{D,a}$.*

Proof. Restoration is a bijection between parts and attained values on each side, and every attained g-value comes from an attained f-value. The kernel of $Q_{D,a}$ on the whole output space is $\{0, (a \text{ at } D)\}$, so each fiber of $Q_{D,a}$ has two elements, and its intersection with $\mathrm{im}\, f$ has at most two. □

Proposition 2.5 (Perfect completeness; Proposition 3.6)**.** *If the source instance is satisfiable, one labeling satisfies every edge.*

Proof. Fix a satisfying assignment ξ of the source variables. At any key, set each retained clause coordinate to the triple given by ξ and each retained bit coordinate to its value under ξ. Label the key by the part containing this point. The rule reads only the key, so it is consistent across all edges that share a vertex. For an edge, both endpoints are labeled by parts containing

the reduction of the same point $x_{\xi} \in \Omega(E)$. The restored outputs are therefore $f(x_{\xi})$ and $g(x_{\xi}) = Q_{D,a} f(x_{\xi})$, and the edge accepts. □

The fibers have size *at most* two. Section 7 pads them to exactly two at a cost of $\delta/3$ in soundness.

2.5 Exact identities forced by joint answers

Shifting an upper row by products of lower rows does not change the joint partition at all. This is "folding" adapted to products.

Lemma 2.6 (Joint folding; Lemma 3.7)**.** *Replace row h of M_D by $(M_D)_h + b_h$, where*

$$b_h = c_h \cdot 1 + \sum_{C,u,v} \lambda_{h,C,u,v} \, (M_C)_u (M_C)_v$$

for proper internal descendants C of D. The new array family has the same key*, and a fixed label's restored outputs change by*

$$\widetilde{y}_C = y_C \ (C \neq D), \qquad (\widetilde{y}_D)_h = (y_D)_h + c_h + \sum \lambda_{h,C,u,v} (y_C)_u (y_C)_v.$$

Proof. By (1) the shift lies in H_D. On the output product, the map that adds the displayed polynomial in the other blocks to block D is an involution, since its inputs are unchanged and the characteristic is two. The new joint function is this bijection composed with the old one. The two therefore have the same equality relation, the same reduced domain and the same partition. Evaluating on any point of a part gives the formula. □

Lemma 2.7 (Joint evaluation; Lemma 3.8)**.** *Fix a legal response with restored outputs y_C and a node D. On*

$$B = \operatorname{span}\big(\{1\} \cup \{(M_C)_u (M_C)_v : C \text{ a proper internal descendant of } D\}\big) \subseteq H_D$$

there is a unique linear functional b_y with $b_y(1) = 1$ and $b_y\big((M_C)_u (M_C)_v\big) = (y_C)_u (y_C)_v$. Every extension $z \in H_D^*$ of b_y has $z(1) = 1$, and

$$\big(z((M_C)_u (M_C)_v)\big)_{u,v} = y_C y_C^{\mathsf{t}}, \qquad \text{a matrix of rank at most one.}$$

Proof. One assignment x realizes the whole response. Point evaluation at x, restricted to B, has the required values and respects every linear relation among the generators. The generators span B, so this fixes b_y. Extend arbitrarily from a basis of B. □

> **Why this is the right invariant.** Lemma 2.7 is the algebraic certificate the soundness proof looks for. An honest answer is evaluation at a point, and evaluation at a point is multiplicative, so its "product form" $F(g,h) = z(gh)$ is a rank-one matrix yy^{t} on every tested list of rows. A cheating prover is forced to produce *some* functional with this property on the rows it was shown. The rest of the proof upgrades "rank one on the tested rows" to "bounded rank everywhere", which decodes to source answers.

2.6 The test distribution

The arrays on the random path are not uniform. They come from a recursive product sampler, which the manuscript calls its own redesign. Only the underlying low-rank bucket viewpoint is credited to Golowich [10], as mediated by Fei–Minzer–Wang [9]. The sampler's structural purpose is this. Every array above a node D_p of the path is a fixed polynomial expression in a bounded number of independent draws at D_p (Lemma 4.1). That bound depends on the row and product counts but not on any branching number.

1. Sample a left source question independently at each leaf, giving E.
2. Sample a uniformly random root-to-leaf path $\mathsf{P} = (D_d, D_{d-1}, \ldots, D_0)$.
3. The scalar sampler $\mathrm{Sample}(D_p)$, at a path node with path child C_*, returns

$$\sum_{C \neq C_*} U_C \; + \; \sum_{h=1}^{R_p} G_h H_h, \qquad U_C \sim \mathrm{Unif}(H_C^2), \quad G_h, H_h \sim \mathrm{Sample}(C_*),$$

   with fresh randomness for every call. At a leaf it returns a uniform element of H_{D_0}.
4. At a path node D_p the array is a sum of *buckets*

$$M_{D_p} = \sum_{v \in K_p \setminus \{0\}} v \otimes h_{p,v}, \qquad h_{p,v} \sim \mathrm{Sample}(D_p) \text{ independently.}$$

   Arrays at nodes off the path are uniform.
5. Choose a level $p \in [d]$ uniformly and a direction $a_p \in K_p \setminus \{0\}$ uniformly. The edge joins the left key of $\Psi_{E,\mathbf{M}}$ to the right key of $Q_{D_p,a_p}\Psi_{E,\mathbf{M}}$.

Each ingredient has a job in the analysis:

- **Products of pairs** make the law of a sampled function have *small bias*, and the bound is independent of dimension (Lemma 3.4). This feeds the Fourier heavy-list argument.
- **Many children** hide which child is on the path. After a cut at D_p is "stopped", the observed data are close in total variation to uniform (Lemma 3.3).
- **Buckets** realize a uniform rank-one step exactly once the cut has been made uniform (Section 3.2). In that law, resampling the bucket h_a changes M by $a \otimes h$ with h uniform, which is the step of the Grassmann/shortcode inverse theorem.
- **Bounded stopped expressions** make the hiding estimates independent of branching. They also let the clean-children argument zero out a predetermined, bounded family of draws.

3 Two toolkits

The soundness proof uses two independent bodies of machinery. The first is the matrix form of the Grassmann/shortcode inverse theorem. The second consists of distributional comparisons for the sampler.

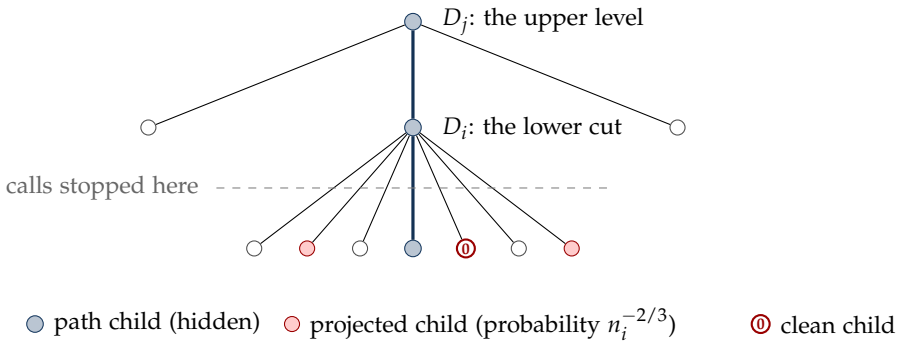


Figure 2: The two levels in the soundness argument. Calls from above that reach D_i are "stopped" there. With many children, the path child is statistically hidden. A sparse random set of children is projected, and a clean child is a projected one all of whose contributions are zero.

3.1 The inverse theorem for matrix tables

Let $K = \mathbb{F}_2^\ell$ and let H be a finite space, so that a matrix $M \in K \otimes H$ is a list of ℓ rows in H. The *rank-one step* takes a uniform M to $M + a \otimes h$, with $a \in K \setminus \{0\}$ and $h \in H$ uniform and independent. Both endpoints are uniform. A *row and column slice* is a nonempty affine set

$$\{M : AM = U,\ Mq_b = t_b\ (b \le c)\},$$

which specifies some row combinations and some column values.

Theorem 3.1 (Inverse theorem; Theorem 2.6)**.** *For every $\eta > 0$ there are $\alpha > 0$ and integers r, ℓ_0 such that for all $\ell \ge \ell_0$, every finite space H, every finite set $\mathcal{Y}$ and every partial table $f : K \otimes H \to \mathcal{Y} \cup \{\perp\}$ the following holds. If*

$$\Pr_{M,a,h}\big[f(M) = f(M + a \otimes h) \neq \perp\big] \ \ge\ \eta,$$

then some value $y \in \mathcal{Y}$ has density at least α on a slice with at most r row and r column constraints. The constants do not depend on $\dim H$ *or* $|\mathcal{Y}|$.

The proof has three short steps on top of the Khot–Minzer–Safra theorem: a set of Grassmann vertices that retains a constant fraction of its edges (so is far from pseudorandom) is dense in an interval of bounded co-order [16] (Theorem 2.3).

1. **Chart** (Lemma 2.4, following [3]). Send M to its graph $L_M = \{(x, M^{\mathrm{t}}x)\} \subseteq \mathbb{F}_2^\ell \oplus \mathbb{F}_2^m$. Then $\dim(L_M \cap L_{M'}) = \ell - \operatorname{rank}(M - M')$, so neighbors in the chart are exactly rank-one changes. A vertex has $(2^\ell - 1)(2^{m+1} - 2)$ Grassmann neighbors: choose one of the $2^\ell - 1$ hyperplanes of L_M, then one of the $2^{m+1} - 2$ lines of the $(m+1)$-dimensional quotient that differ from the image of L_M. Of these, $(2^\ell - 1)(2^m - 1)$ stay in the chart, *exactly half*. Retention therefore drops by at most a factor of two. Containment of A_0 and containment in B_0 translate into row and column equations.
2. **Dummy columns** (Lemma 2.5). The KMS theorem needs large ambient dimension, but H may be tiny, even zero. Pad with dummy columns. A slice in the padded space projects to a slice in the original one, and *every* point has the same number of lifts, so densities are preserved exactly.

3. **Many values** (Theorem 2.6). With μ_y the mass of the fiber $f^{-1}(y)$ and θ_y its retention, the defined-equality probability equals $\sum_y \mu_y \theta_y$. Since $\sum_y \mu_y \le 1$, some single fiber has $\theta_y \ge \eta$. No factor $|\mathcal{Y}|$ is lost; this averaging is also in [3]. The chart is from [3] as well. The dummy-column elimination and the many-fibers lemma below are the manuscript's own.

The independence from $|\mathcal{Y}|$ is essential: the upper table's responses record every lower block, so its range is enormous. The analysis also needs a slice that a player can *find* from random row advice.

Lemma 3.2 (Many good row fibers; Lemma 2.7)**.** *Let α, r be the constants of Theorem 3.1 at threshold $\eta/2$. If a partial table has defined-equality probability at least η, then a uniform $A \in \mathrm{Hom}(K, \mathbb{F}_2^r)$ and an independent uniform M satisfy: with probability at least $(\eta/4)2^{-r\ell}$, the row fiber $\{M' : AM' = AM\}$ contains a slice (at most r further column constraints) on which a single value has density α. The witness is a deterministic function of (f, A, AM).*

Proof idea. Erase f on the union T of all good row fibers. If T had measure below $\eta/4$, the erased table would still have defined equality above $\eta/2$, so the inverse theorem would produce a good fiber of the erased table. That fiber lies in T, where the erased table is undefined, a contradiction. Average over the $2^{r\ell}$ row maps. □

3.2 Comparisons for the sampler

The analysis repeatedly replaces the true sampling law by simpler ones. Each replacement is justified for a specified *observation*, and the bookkeeping of what may be observed is the subtle part.

Stopping at a cut (Lemma 4.1). Fix the path down to a node D_p. Each scalar call above D_p is a sum of uniform functions from ordinary children and R_h products of two fresh calls to the path child. Stop every call when it reaches D_p. Every ancestor array is then a fixed polynomial expression in an *exterior record* and at most

$$T_p = \sum_{k=p}^{d} (2^{\ell_k} - 1) \prod_{h=p+1}^{k} 2R_h$$

pending draws at D_p. Crucially, T_p does not depend on any branching number.

One hidden child (Lemma 4.2). Each draw at D_p and each array below it decomposes into independent *child blocks* X_C, one per child C of D_p. Under the true law, the block of the (unknown) path child J has some law Q_C, while every other block is uniform (P_C). Each block lives in a space whose dimension depends only on lower parameters, so the density $q_C = dQ_C/dP_C$ is at most a constant L_p.

Lemma 3.3. *Forgetting the path below D_p, the true law of the blocks is within total variation $L_p/(2\sqrt{n_p})$ of the uniform law.*

Proof. The true density with respect to $P = \prod_C P_C$ is $Z = \frac{1}{n_p}\sum_C q_C(X_C)$, an average of independent mean-one variables. Then $\mathbb{E}_P(Z-1)^2 = n_p^{-2}\sum_C \mathbb{E}(q_C-1)^2 \le L_p^2/n_p$, and $\mathrm{TV} \le \frac{1}{2}\sqrt{\chi^2}$. □

The bound is lost if the observation reveals J, and much of the care in the manuscript goes into never revealing it. In the uniform law the array at D_p is uniform, and resampling one bucket h_a gives *exactly* the uniform rank-one step $M \mapsto M + a \otimes (h_a + h'_a)$.

Sparse projection (Lemma 4.3). At a lower cut D_i, mark each child independently with probability $\beta_i = n_i^{-2/3}$. At every leaf of a marked child, replace the clause question by a variable question, sampled conditionally as in a source edge. Make all draws at the cut uniform in the *projected* spaces. Viewed through the pullback to the original domains, the resulting *modified* law is close to uniform:

$$1 + \chi^2 = \prod_C \big(1 + \beta_i^2 \chi^2(Q_C \| P_C)\big) \le (1 + \beta_i^2 L_i^2)^{n_i} \le \exp(L_i^2 n_i^{-1/3}).$$

This is a form of the Khot–Safra sub-code covering idea [17].

Posterior of a selected child (Lemma 4.4). Bayes' rule gives, for every child and every observed lifted data,

$$\Pr(C \text{ marked} \mid \text{data}) = \frac{\beta_i q_C}{1 - \beta_i + \beta_i q_C} \le \frac{\beta_i L_i}{1 - \beta_i} =: q_i.$$

This holds even for a child chosen *after* looking at the data. A useful consequence: if two bilinear forms on $H_{D_i}(E)$ are determined by the lifted data and differ, a witness pair lies in the square spaces of at most two children. If neither child is marked, their restrictions to the projected space also differ. So

$$\Pr\big(F_0 \ne F_1,\ F_0|_{H'} = F_1|_{H'}\big) \le 2q_i.$$

Small bias independent of dimension (Lemma 4.5). Call a law on a space *quarter-balanced* if every nonzero linear functional takes each value with probability at least $1/4$.

Lemma 3.4. *If g, h are independent draws from quarter-balanced laws and B is a nonzero bilinear form, then $|\mathbb{E}(-1)^{B(g,h)}| \le 7/8$. Consequently, if every $R_p \ge R_{\min}$ (an absolute constant), every scalar law is quarter-balanced, and every nonzero character has bias at most $(7/8)^{R_p}$ at level p.*

Proof. The kernel of $g \mapsto B(g,\cdot)$ is a proper subspace, so some nonzero functional vanishes on it, and quarter balance gives $\Pr(B(g,\cdot) \ne 0) \ge 1/4$. Given such g, the bit $B(g,h)$ takes each value with probability at least $1/4$. Each value of $B(g,h)$ therefore has probability at least $1/16$, and the bias is at most $7/8$.

For the induction, let $f \ne 0$ on H_{D_p}. If f is nonzero on an ordinary child's square space, the uniform summand makes $f(\cdot)$ an unbiased bit. Otherwise f is nonzero on the path child's square space, so $B(g,h) = f(gh)$ is a nonzero bilinear form there. Each of the R_p independent products then contributes a factor at most $7/8$. □

The bound uses only whether certain maps are nonzero, so no dimension enters. This is why the Fourier argument of Section 4 survives function spaces whose dimension grows with the source alphabet.

4 Soundness I: two levels and an upper decoder

Fix a labeling that, on a NO instance, accepts with probability at least ε. The goal of this and the next two sections is a contradiction.

4.1 Two levels that succeed together

Let Acc_p be acceptance of the test at level p, with all directions sampled at once, and let $N = \sum_p \mathbf{1}_{\mathrm{Acc}_p}$. By Jensen, $\mathbb{E}[N(N-1)] \geq (\mathbb{E}N)^2 - \mathbb{E}N \geq d^2\varepsilon^2 - d\varepsilon$. If $d\varepsilon \geq 4$, averaging over the $d(d-1)$ ordered pairs gives levels $i < j$ with

$$\Pr(\mathrm{Acc}_i \cap \mathrm{Acc}_j) \;\geq\; \varepsilon^2 - \varepsilon/d \;\geq\; \tfrac{3}{4}\varepsilon^2.$$

Pass to the *modified* law at level i, in which children of D_i are sparsely projected and draws at D_i are uniform. With branching large, this costs at most $\varepsilon^2/4$, leaving $\Pr_i(\mathrm{Acc}_i \cap \mathrm{Acc}_j) \geq c := \varepsilon^2/2$.

Two levels are needed because each level supplies something the other lacks. Success at the *upper* level j gives the inverse-theorem structure. Success at the *lower* level i ties that structure to the lower rows, on which joint legality certifies rank one, and to the projected children, where the right player lives.

4.2 The upper table

Let $H = H_{D_j}(E)$ and $M = M_{D_j}$. The inverse theorem needs a table on $K \otimes (\text{something})$ under the rank-one step. The difficulty is that the response at M depends on lower rows, and the analysis must not reveal which level-i node is on the path. The manuscript therefore lets

$$B = \mathrm{span}\big(\{1\} \cup \{s_{C,t}s_{C,u} : C \text{ any level-}i \text{ descendant of } D_j\}\big) \subseteq H$$

include the products of displayed rows at *every* level-i descendant, and works on H/B. Choose a linear section $s : H/B \to H$. The table $f_P(\bar{M})$ queries the labeling with upper array $s(\bar{M})$ and records the whole response: the upper output u and every lower output y_C. Lemma 2.6 recovers the response at the true M, because $M - s(\bar{M})$ has rows in B. The upper output becomes $u + (M - s(\bar{M}))b_y$, and the lower outputs are unchanged. Responses where level-i success has conditional probability below $c/2$ are marked undefined.

Lemma 4.1 (Lemma 5.1)**.** *Resampling the bucket of M in the omitted direction a produces two arrays $M, M' = M + a \otimes h$. The partial table has defined equality at least $c^2/8$ on them, and at least $\theta = c^2/16$ under the uniform rank-one step on $K \otimes (H/B)$.*

Why. The right question at level j sees only M modulo $\langle a \rangle$, so it is the same for both draws. Successful left responses therefore agree with it, and up to the folding correction, which also moves only along a, there are at most two possible defined responses. Two independent draws coincide with probability at least $(p_1 + p_2)^2/2$, and Cauchy–Schwarz finishes. In the uniform law at the upper cut, bucket resampling is exactly the rank-one step. □

4.3 The coset of functionals

Put $\eta = \theta/2 = c^2/32$, so that with probability at least η over the lower data the table has defined equality at least η. Apply Lemma 3.2 with random row advice $A \in \mathrm{Hom}(K, \mathbb{F}_2^r)$. With probability at least $g = (\eta^2/4)2^{-r\ell}$ the advice is good. Its witness consists of at most r column functionals $q_a \in (H/B)^*$, values v_a, and a response $t_* = (u, (y_C)_C)$ that occurs with density α on the slice. Pull the q_a back to H and let $Z = \mathrm{span}\{q_a\} \subseteq \mathrm{Ann}(B)$. The response t_* defines b_y on B as in Lemma 2.7. Define $z_0 \in H^*$ to be b_y on B and 0 on $s(H/B)$. At every match the restored upper output is $Mz_0 + u$, and *every* z in the coset $z_0 + Z$ satisfies

$$z(1) = 1, \qquad \big(z(s_{C,t}s_{C,u})\big)_{t,u} = y_C y_C^{\mathsf{t}} \quad \text{for every level-}i\text{ descendant } C.$$

So the left player has a whole coset of normalized functionals, each with tested rank at most one at every lower node. Lemma 5.2 records the quantitative form. Let J be the event "good advice and the column values match". Then

$$\Pr_i\big(J,\ R_j = Mz_0 + u\big) - \rho \Pr_i(J) \ \geq\ b, \qquad \rho = \kappa\alpha/4, \quad b = \kappa\alpha q_0/4,$$

where R_j is the level-j block restored from the right answer of the level-i test, and $q_0 = g2^{-r\ell}$. The difference form matters. Disintegrating it over the exposed data gives, with probability at least b, a slice of positive mass on which the conditional prediction probability is at least ρ. That is exactly the hypothesis of the heavy-list lemma.

4.4 What each player sees

The proof now builds two *local* decoders: a left one that sees only left-player information, and a right one that sees only right-player information. Split $M = M_{\mathrm{known}} + X$, where $X = \sum_{v \in \ker A \setminus 0} v \otimes h_v$ collects the buckets that the row advice cannot see.

- The **left input** $\mathcal{L}$ is E, the path through D_i, A, all other arrays (including the full lower array $S = M_{D_i}$), and the known buckets. It does *not* include the omitted lower direction a_i.
- The **right input** $\mathcal{R}$ is O, the path, A, a_i, the other arrays on projected domains, the known buckets, and S only modulo a_i.
- Neither player sees X. Its law $\mu_{\mathcal{R}}$ is computable from the right input alone, and it is *exactly* the conditional law given both inputs, because hidden buckets use fresh, disjoint randomness. By Lemma 3.4, every nonzero character of X has bias at most $\lambda = (7/8)^{R_j}$.

The right player can evaluate $G_{\mathcal{R}}(X)$, the restored upper response of its own level-i query when the hidden part equals X.

4.5 A heavy list that works for every witness

Lemma 4.2 (One heavy list; Lemma 5.3)**.** *Let μ be a law on $V = W \otimes H'$ in which every nonzero character has bias at most λ, where* $\operatorname{codim} W \leq r$*. Let $G : V \to K$, let $h_0 = 2^{-r\ell}$ and $\tau = \rho h_0/4$, and suppose $2^{-(\ell-r)} < \rho/8$ and $\lambda \leq \min(h_0/2, \tau^2/2)$. For $\sigma \in K^*$ let*

$$\mathcal{H}_\sigma = \big\{\Phi \in V^* : |\mathbb{E}_\mu[(-1)^{\sigma(G(X))}\chi_\Phi(X)]| \geq \tau\big\}.$$

Then $|\mathcal{H}_\sigma| \le 2/\tau^2$. *Moreover, if on an affine slice* $\mathcal{C} = \{X|_Q = t_Q\}$ *with* $\dim Q \le r$ *one has* $\Pr_\mu(G(X) = Xz' + k_0 \mid \mathcal{C}) \ge \rho$, *then for some* σ *with* $\sigma|_W \neq 0$ *the list* $\mathcal{H}_\sigma$ *meets* $(\sigma|_W) \otimes z' + W^* \otimes Q$.

Proof sketch. The slice indicator is an average of $|W^* \otimes Q|$ characters, and small bias gives $\mu(\mathcal{C}) \ge h_0/2$. Expanding the event $G(X) = Xz' + k_0$ over $\sigma \in K^*$, at most a $\rho/8$ fraction of the σ annihilate W. Some other σ therefore correlates with the character $(\sigma|_W) \otimes z'$ on the slice. Removing the slice indicator by its character expansion leaves an unconditional correlation of at least τ with a character in the stated coset. For the size bound, sum m sign-aligned heavy characters and apply Cauchy–Schwarz: $m^2\tau^2 \le m + m^2\lambda$, using only that products of distinct characters have bias at most λ. □

The lists depend only on the right player's input, not on the left player's witness. That is what makes the decoding local.

Proposition 4.3 (Upper decoding; Proposition 5.4)**.** *There are randomized local strategies, a left one on* $\mathcal{L}$ *returning a bilinear form* F_L *on* $H_i = H_{D_i}(E)$ *and a right one on* $\mathcal{R}$ *returning* F_R *on* $H_i' = H_{D_i}(O)$, *such that*

$$\Pr\big(F_R = F_L|_{H_i' \times H_i'}\big) \;\ge\; \gamma_j := b\, 2^{-\ell_j} \frac{\tau^2}{2}\, 2^{-r} > 0.$$

Every left output has the form $F_L(x,y) = z(xy)$ *with* $z(1) = 1$ *and* $\operatorname{rank}\big(F_L(S_t, S_u)\big)_{t,u} \le 1$. *The constant* γ_j *does not depend on the lower row count* ℓ_i, *on any branching number, or on the source alphabet.*

The two decoders. The *left* decoder samples z uniformly from the coset $z_0 + Z$ and outputs $F_L(x,y) = z(xy)$. On bad advice it outputs point evaluation at a fixed legal assignment. The *right* decoder picks $\sigma \in K^*$ at random and a random Φ from $\mathcal{H}_\sigma$. It fixes $h_\sigma \in W$ with $\sigma(h_\sigma) = 1$, sets $z_R = \Phi(h_\sigma)$, and outputs $F_R(x,y) = z_R(xy)$. When the witness is favorable, some list element lies in $(\sigma|_W) \otimes z' + W^* \otimes Q$. Contracting it with h_σ lands in $z' + Q$, the restriction of the coset to the projected space, and the left's uniform choice hits it with probability $2^{-\dim Q} \ge 2^{-r}$.

> **Order of quantifiers.** The constant γ_j depends only on ε, the inverse-theorem constants and ℓ_j. It does not depend on the lower row count ℓ_i. This is what lets the parameters be chosen top down: first ℓ_j, then γ_j, then a lower row count ℓ_i large enough for the rank test of Section 5.

5 Soundness II: from tested rank to full rank

The left form $F_L(x,y) = z(xy)$ has rank at most one *on the displayed lower rows* $S_1, \dots, S_{\ell_i}$. That alone says little, because the left decoder chose z after seeing those rows. A form can be rank one on the rows it was built around and still have huge rank. This section shows that a positive fraction of the agreement nevertheless comes from forms of bounded *full* rank. The manuscript notes that Fei, Minzer and Wang have a close analogue of this step [9].

Call F *admissible* at S if $F(x,y) = z(xy)$ with $z(1) = 1$ and $\operatorname{rank}\operatorname{Gram}_F(S) \le 1$, where $\operatorname{Gram}_F(S) = (F(S_u, S_v))_{u,v}$.

5.1 Resampling in the lower fiber

Suppose that high-rank agreement, with $\operatorname{rank} F_L > s$, has probability h. Fix both players' random tapes, which preserves this probability, and erase every left output that is not admissible or has rank at most s. The argument has three steps.

1. **Same right view, two left views.** The right player sees S only modulo a_i. Draw S and S' independently from the same fiber S mod a_i. The right answer is the same for both, so both left forms restrict to it with probability $\mathbb{E}p^2 \geq (\mathbb{E}p)^2 = h^2$. As an unfiltered law, the pair is just $S' = S + a_i \otimes l$ with S, l uniform.
2. **Restrictions equal ⇒ forms equal.** Both full left forms are functions of the lifted data. By the posterior bound (Section 3.2), unequal forms with equal restrictions to the projected space have probability at most $2q_i$.
3. **Transfer to uniform.** The remaining event, full defined equality $f(S) = f(S + a_i \otimes l)$, is a function of lifted data. It therefore transfers to the unrestricted uniform law at a total-variation cost v_i.

The result is a partial table f on $K_i \otimes H_i$ under the uniform rank-one step, with defined equality at least $h^2 - 2q_i - v_i$ (Lemma 6.1). Every defined output is admissible at its own argument and has rank greater than s.

5.2 High-rank forms fail the rank test on slices

Apply Theorem 3.1 to this table. It returns one *fixed* form F that is the output on an α'-dense part of an affine row/column slice. Every such occurrence has $\operatorname{rank} F > s$ and $\operatorname{rank} \operatorname{Gram}_F(S) \leq 1$. The next lemma says this is impossible. It is elementary and pleasant.

Lemma 5.1 (Lemma 6.2)**.** *Let F be a bilinear form on H, and let $\mathcal{A} \subseteq H^\ell$ be a nonempty slice with at most r' row constraints and at most r' column constraints. If $\ell \geq 2m + r'$, then for uniform $S \in \mathcal{A}$*

$$\Pr\big[\operatorname{rank} \operatorname{Gram}_F(S) \leq 1\big] \;\leq\; (2^m - 1)\, 2^{-(\operatorname{rank} F - 2r')} + \big[1 + (2^m - 1)^2\big] 2^{-m^2}.$$

Proof. An invertible change of rows $Y = TS$ turns the row constraints into "the first $a \leq r'$ rows are fixed". It does not change the rank of the Gram matrix, since $\operatorname{Gram}_F(Y) = T \operatorname{Gram}_F(S) T^{\mathsf{t}}$. The column constraints say each free row lies in a translate of $N = \bigcap \ker q_v$, a subspace of codimension $c \leq r'$, and the free rows are independent. Restricting F to $N \times N$ loses at most $2c$ in rank, so $t := \operatorname{rank} F|_N \geq \operatorname{rank} F - 2r'$.

Take two disjoint groups of m free rows, $x_1, \ldots, x_m$ and $y_1, \ldots, y_m$. The functionals $\lambda_u = F(x_u, \cdot)|_N$ are independent, each uniform on a translate of the t-dimensional image of $F|_N$. A nonzero combination $\sum e_u \lambda_u$ is uniform on such a translate, hence zero with probability at most 2^{-t}. A union bound over the $2^m - 1$ combinations shows the λ_u are linearly independent except with probability $(2^m - 1)2^{-t}$. When they are independent, each column $(F(x_u, y_v))_u$ is uniform in $\mathbb{F}_2^m$. The $m \times m$ cross matrix is then uniform, and exactly $1 + (2^m - 1)^2$ of the 2^{m^2} matrices have rank at most one. A rank-≤ 1 Gram matrix has a rank-≤ 1 cross submatrix. □

Choose m so that the second term is below $\alpha'/2$, then s so that $2^{m-(s-2r')} < \alpha'/2$, then $\ell_i \geq 2m + r'$ (and at least the inverse theorem's ℓ_0'). The fixed form F then passes the rank test on less than an α' fraction of the slice, contradicting its density.

Proposition 5.2 (Low-rank agreement; Proposition 6.3)**.** *With these choices and n_i large,*

$$\Pr\big[F_L|_{H_i'} = F_R,\ F_L \text{ admissible},\ \operatorname{rank} F_L \le s\big] \ \ge\ \gamma_j/2.$$

The cutoff s and the row count ℓ_i depend only on γ_j.

> **What was gained.** Before this step the left form was certified only on the rows the verifier displayed. Now, with probability $\gamma_j/2$, it is a normalized product form of rank at most s on the *entire* space H_i, and it agrees after projection with a form the right player computed on its own. A bounded rank on all of H_i is what lets the form be read at individual leaves.

6 Soundness III: odd lists, clean children, and repetition

It remains to turn bounded-rank agreement into a strategy for the source game, and to show that it wins too often.

6.1 Reading a form at a leaf

For a finite set Λ, let $\mathcal{O}_s(\Lambda)$ be the subsets of odd size at most s. For a map $\pi : \Lambda \to \Gamma$, the *parity pushforward* is

$$\pi_{\mathrm{odd}}(L) = \{b \in \Gamma :\ |L \cap \pi^{-1}(b)| \text{ is odd}\}.$$

The *odd-list game* $G_{\mathrm{odd},s}$ is the source game with answers in $\mathcal{O}_s$ and constraints π_{odd}. The same pushforward describes how Fourier characters restrict in low-degree long-code analysis (Dinur–Guruswami).

Lemma 6.1 (Lemma 7.1)**.** 1. *π_{odd} maps $\mathcal{O}_s(\Lambda)$ into $\mathcal{O}_s(\Gamma)$, and* $\operatorname{val}(G_{\mathrm{odd},s}) \le s^2 \operatorname{val}(G_{\mathrm{src}})$.
2. *If $F(x,y) = z(xy)$ on H_i with $z(1) = 1$ and* $\operatorname{rank} F \le s$*, then at every leaf λ below D_i the set $\{a : z(e_a) = 1\}$ is in $\mathcal{O}_s(\Lambda_\lambda)$, where e_a is the indicator that the answer at λ is a.*
3. *If the leaf is projected by π_e and $F' = F|_{H_i'}$, the corresponding set for F' is the parity pushforward of the left set.*

Proof. (1) The sizes satisfy $|\pi_{\mathrm{odd}}(L)| \equiv |L| \equiv 1 \pmod 2$, and each element of $\pi_{\mathrm{odd}}(L)$ has its own nonempty, disjoint preimage in L, so $|\pi_{\mathrm{odd}}(L)| \le |L|$. For the value bound, let each player answer with a uniformly random element of its list. On an accepted list edge, every right element has an odd, hence positive, number of preimages in the left list. So some pair accepts with probability at least s^{-2}.

(2) The indicators e_a lie in the leaf space and satisfy $e_a e_b = 0$ for $a \ne b$, $e_a^2 = e_a$, and $\sum_a e_a = 1$. Hence $(F(e_a, e_b))$ is diagonal with entries $d_a = z(e_a)$. Its rank, the number of a with $d_a = 1$, is at most s, and $\sum_a d_a = z(1) = 1$, so that number is odd.

(3) The pullback of the right indicator e'_b is $\sum_{\pi_e(a)=b} e_a$, so $F'(e'_b, e'_b) = \sum_{\pi_e(a)=b} d_a$. □

This is the clean replacement for the usual "decode a long code to a short list" step. The orthogonal idempotents e_a diagonalize every product form, and normalization $z(1) = 1$ forces the decoded list to be odd and hence nonempty.

6.2 Clean children

To apply parallel repetition, the proof needs *independent* source edges together with *local* rules that answer them. The trouble is that each player's input involves sampled functions that depend on all leaves. The manuscript's device, which it traces to the zero-advice idea of Khot–Minzer–Safra [15, Remark 3.2], is to find children whose contributions vanish.

Write every one of the T_i^{cl} draws at the cut D_i (the rows of S together with all stopped calls) as a sum of independent child contributions $U_{q,C} \in H_C(O)^2$. A child C is *clean* if it is projected, all its contributions $U_{q,C}$ are zero, and every array in its subtree is zero.

Lemma 6.2 (Exact clean law; Lemma 7.2)**.** *Given the exterior data, the clean indicators are independent, each with probability exactly $\beta_i p_{0,i}$, where*

$$p_{0,i} = 2^{-T_i^{\mathrm{cl}} \dim(H_C'^2) - \sum_{D \preceq C} \ell_{\mathrm{level}(D)} \dim H_D'}$$

is the same for every child and every choice of source questions*. Conditional on all public data, the source edges at the designated (first) leaves of the clean children are independent, each with the source distribution μ.*

Why the law is exact. A fully projected subtree has answer domain $\mathbb{F}_2^t$ at every leaf, whatever the variable IDs are. Its function spaces, and hence their dimensions, depend only on the shape of the subtree. The probability that a fixed number of independent uniform vectors all vanish is therefore the same for every question tuple. The constant factor cancels when conditioning on cleanliness, so the edge tuple in a clean child keeps its product law. □

> **Why exactness matters.** If the probability of being clean depended, even slightly, on the source questions, conditioning on cleanliness would bias those questions, and the repetition theorem would no longer apply. A merely positive lower bound would not do; the manuscript stresses this point. Two facts make the probability exactly constant. First, asking each child's contributions to vanish separately, rather than their sum, makes cleanliness an event about that child's own latent summands. These events are independent across children, and cancellations or shared constants cannot affect them. Second, a fully projected subtree has function spaces that depend only on its shape, because repeated variable IDs still occupy separate coordinates. The probability is therefore a dimension count that is the same for every question tuple.

Lemma 6.3 (Local reconstruction; Lemma 7.3)**.** *After the public data are fixed, each player can rebuild its exact decoder input from its own questions at the designated clean leaves. The left player recovers E from its clause questions; the right player recovers O from its variable questions.*

The reason is that every sampled function ignores the arguments in clean children. Clean contributions are zero, nonclean contributions live elsewhere, and the functions that ignore those arguments form an algebra. Canonicalization then discards the clean slots. The left decoder is run without a_i and the right decoder with S only modulo a_i, exactly as specified. The right player's heavy lists are computed from the fixed reference law $\mu_{\mathcal{R}}$, so no small-bias claim is needed after conditioning on cleanliness.

6.3 The repetition bound

Proposition 6.4 (Proposition 7.4). *If* $\mathrm{val}(G_{\mathrm{src}}) < 1/(2s^2)$*, then the bounded-rank agreement event* $\mathcal{L}_{i,j,s}$ *of Proposition 5.2 satisfies*

$$\Pr(\mathcal{L}_{i,j,s}) \;\le\; \Big(1 - \frac{\beta_i p_{0,i}}{64}\Big)^{n_i} \;\le\; \exp\Big(-\frac{p_{0,i}}{64}\, n_i^{1/3}\Big).$$

Proof. By Lemma 6.1, $\mathrm{val}(G_{\mathrm{odd},s}) < 1/2$. Conditional on the public data with k clean children, the reconstructed decoders, followed by reading diagonal supports at the designated leaves, form local strategies for $G_{\mathrm{odd},s}^{\otimes k}$. Whenever $\mathcal{L}_{i,j,s}$ occurs, they win every coordinate. Dinur–Steurer with gap $1/2$ bounds this by $(1 - \frac{1}{64})^k$. Averaging $(63/64)^{|C|}$ over independent clean indicators gives $\prod_C (1 - \beta_i p_{0,i} + \frac{63}{64}\beta_i p_{0,i}) = (1 - \beta_i p_{0,i}/64)^{n_i}$, and $n_i \beta_i = n_i^{1/3}$. □

The constant $p_{0,i}$ depends on the source power t, the row and product counts, and branching *below* level i, but not on n_i. Taking $n_i^{1/3} > \frac{64}{p_{0,i}} \log \frac{4}{\gamma_j}$ makes the bound smaller than $\gamma_j/4$, contradicting Proposition 5.2.

7 Assembly: parameters, exact fibers, and weights

7.1 The order of choices

Every estimate above is uniform in the quantities chosen after it. The order is the following (Section 8 of the manuscript):

1. From δ: put $\varepsilon = \delta/3$, choose depth d with $d\varepsilon \ge 4$, and set $c = \varepsilon^2/2$, $\kappa = c/2$, $\eta = c^2/32$. Take the inverse-theorem constants α, r, ℓ_0 at threshold $\eta/2$.
2. **Row counts, top down.** At level j, choose ℓ_j large enough for every lower rank test required by levels above j, and with $2^{-(\ell_j - r)} < \rho/8$. This determines γ_j. Then choose the product count R_j with $(7/8)^{R_j} \le \lambda_j$, above an absolute minimum.
3. **Rank cutoff.** Let s be the largest of the cutoffs from Section 5.
4. **Source power.** Choose t with $\mathrm{val}(G_{\mathrm{src}}) < 1/(2s^2)$. This fixes the alphabets. It changes the dimension of every function space, but no earlier constant depends on dimension: that is the point of the dimension-free inverse theorem, heavy list and bias bound.
5. **Branching, bottom up.** Choose cubes $n_1, n_2, \ldots, n_d$ one at a time, in ascending order of level. At level p, the likelihood bound L_p and the clean probability $p_{0,p}$ depend only on what was fixed earlier. Large n_p therefore makes all of $L_p/\sqrt{n_p}$, the sparse-projection error, the posterior q_p, and the clean bound $\exp(-p_{0,p} n_p^{1/3}/64)$ as small as needed. Higher branching only adds exterior data, so it cannot spoil an earlier choice.

With these choices the preliminary game has perfect completeness, fibers of size at most two, and value below ε on NO instances (Proposition 8.1). Otherwise Section 4 produces levels $i < j$ with agreement γ_j, Section 5 reduces it to bounded-rank agreement $\gamma_j/2$, and Section 6 bounds that by less than $\gamma_j/4$.

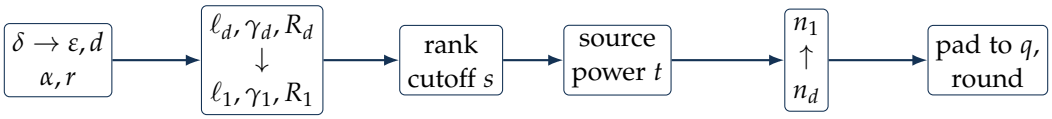


Figure 3: The order in which the constants are fixed. Each estimate is uniform in everything to its right.

7.2 Exactly two preimages

The constraints so far are *at most* 2-to-1 on legal answers. Embed all legal answers in common alphabets $[2q]$ and $[q]$, where q is large compared with the constant bounds on legal answers and with $1/\delta$. For an edge with legal map $\pi : L \to R$, every right label b has residual capacity $c_b = 2 - |\pi^{-1}(b)|$. The $N = 2q - |L|$ illegal left labels are distributed into these capacities in *all* possible ways, each completion getting equal weight. Every completion is exactly 2-to-1 and agrees with π on legal answers, so perfect completeness survives edge by edge. By symmetry, a fixed illegal left label maps to a fixed right label with probability $c_b/N \le 2/N \le \delta/3$. Illegal answers can therefore add at most $\delta/3$ to the value.

7.3 Removing weights

All weights are rational with polynomial bit length. With m weighted entries, set $K = \lceil 3m/\delta \rceil$ and replace weight w_a by $\lfloor K w_a \rfloor$ copies, plus one copy for each of the first few entries to bring the total to K. Only edges already in the support are used, so completeness stays 1. The total variation is at most $m/K \le \delta/3$. The final soundness is $\varepsilon + \delta/3 + \delta/3 = \delta$.

8 Reader's guide

What is external. Three theorems are used as black boxes:

- Dinur's PCP with perfect completeness [6];
- the Dinur–Steurer bound $\mathrm{val}(G^{\otimes k}) \le (1 - g^2/16)^k$ for projection games with value at most $1 - g$ [8];
- the Khot–Minzer–Safra theorem that pseudorandom sets in the Grassmann graph expand near-perfectly [16].

Everything else is proved in the manuscript, including the extensions of the inverse theorem. The chart and the fiber averaging follow Barak–Kothari–Steurer [3]; the dummy-column and many-fibers lemmas are new.

What is new relative to the 2-to-2 theorem. A remark of mine, not the manuscript's: the 2-to-2 reduction is built on linear equations over $\mathbb{F}_2$, and satisfiable linear systems can be solved by Gaussian elimination. That route therefore cannot have perfect completeness unless $\mathsf{P} = \mathsf{NP}$, and repetition is unavailable for 2-to-1 games. The manuscript instead starts from a satisfiable CSP and uses a test that an honest prover passes *exactly*: questions are unlabeled partitions

given by joint evaluation of a tree of polynomial arrays, and one output direction is erased. Soundness then rests on four mechanisms:

1. exact algebraic identities forced by joint legality;
2. a hidden-path sampler whose statistics are uniform in all dimensions;
3. a rank-promotion step, from tested rank one to bounded full rank;
4. an exactly balanced "clean child" event that recreates independent source edges.

The manuscript credits Golowich [10] and Fei–Minzer–Wang [9] for the bucket viewpoint and for an analogue of the rank step.

Where to look first. The proof is long but modular, and its information bookkeeping is unusually explicit. A referee should concentrate on four places:

- **The information sets.** Section 5.3 of the manuscript defines exactly what each decoder sees. The left decoder never receives the omitted lower direction, and the right decoder receives the lower array only modulo that direction. These restrictions are used again in the reconstruction of Lemma 7.3, and the two descriptions must match.
- **The comparison chain.** Each total-variation step (original $\to$ modified at the lower cut $\to$ uniform at the upper cut) is valid only for observations that do not reveal the hidden path or the projection marks. The "common observations" paragraph in Section 4.3 states what may be observed. Check that every event to which a comparison is applied is of that kind.
- **Disintegrations.** The proof is careful not to condition on success before applying the posterior bound or the bias bound. Lemma 5.2 and Proposition 5.4 disintegrate an unconditional inequality instead. This is the point most likely to hide a slip.
- **Exactness of the clean law.** Lemma 7.2 needs the cleanliness probability to be literally independent of the source questions, which holds because projected subtrees have identical function spaces. Check that the padded call family really is fixed before the row advice selects visible buckets.

Formal verification. The result comes with a Lean formalization: the Comparator challenge `PerfectCompleteness.lean`, whose solution is a development of about 580 Lean files, plus shared libraries, in the repository's `lean/` directory. The challenge permits only Lean's standard axioms (`propext`, `Quot.sound`, `Classical.choice`), so none of the external theorems is assumed. The formal route need not follow the paper's citations: for example, its matrix inverse theorem is imported from the release's Unique Games development. If the check passes, the remaining question is whether the formal statement matches Theorem 1.1. I have not run the check.

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